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Application of an alluvial architecture model to better understand seepage risk in floodplains

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ABSTRACT

The amount and distribution of coarse-grained sediment (e.g., sands and gravels) relative to fine-grained sediment (e.g., clays and silts) within a floodplain influences many of the floodplain geotechnical properties, including the potential for groundwater seepage. Seepage is a primary driver of levee and dam failure, and understanding its potential is of paramount concern to engineers and resource managers. This paper reports the results of a computational modeling study that simulated alluvial floodplain construction using a suite of simple geomorphic process-imitating rules.

A model aggrades a floodplain cross section within an alluvial basin, creating floodplain architecture by differentiating between sediment deposited by channel processes (sand) and sediment deposited by overbank flood processes (clay). The evolution of two floodplain cross sections of the Trinity River, near Dallas, Texas is simulated using five different experimental scenarios. The study area is the site of large levee rehabilitation projects in which accurate characterization of the geologic environment has significant engineering importance. Study results predict that scenario components including the alluvial basin width, the initial topography of the floodplain base level, and the channel aggradation rate significantly affect the fraction of the floodplain width that contains channel deposits by influencing the avulsion frequency of the river during floodplain construction. Increased avulsion frequency equated to more numerous, yet smaller channel deposits. The dimensions of the channel deposits predicted by this study are similar to those typically observed in large, fully meandering river systems. The model devised for this study is relatively simple and can be run in multiple iterations to produce probabilistic outputs, such as the likely range of channel deposit widths within a floodplain cross section. This type of information is useful to engineers for a host of applications including predicting the data collection density necessary to characterize the geotechnical properties of a project site.

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1. Introduction

The alluvial architecture of a floodplain refers to the lithology and spatial distribution of the sedimentary facies that compose the floodplain subsurface (Allan, 1978; Shanley and McCabe, 1993). The character of the alluvial architecture plays an important role in determining the geotechnical properties within the floodplain, such as areas of high and low fluid seepage potential (Webb and Davis, 1998; Willis and Tang, 2010; Li and Caers, 2011). Relatively coarse-grained architectural elements, such as sand or gravel sedimentary deposits, are porous and can serve as effective reservoirs of underground fluid, such as water and hydrocarbon. Also, coarse-grained deposits have relatively high hydraulic conductivities and can serve as efficient seepage conduits. Because of these properties, it is often necessary to understand the alluvial architecture of an area to properly construct and maintain engineering structures that might become adversely affected by subsurface seepage, such as dams or levees (May and Schmitz, 1996). Identifying floodplain areas with high seepage potential due to the underlying lithology (i.e., coarse-grained sediment deposits) offers engineers the opportunity to modify project plans and take precautions to reduce the probability that the engineered structure will become adversely impacted from seepage.

The alluvial architecture of an individual floodplain is difficult to define precisely because of the large volume of floodplain material involved, the architectural elements occur in a wide range of sizes, and most of the elements occur in the subsurface and are not directly measurable (Collinson, 1978; Allen, 1979; Koltermann and Gorelick, 1996). However, the distribution of the sedimentary deposits composing a floodplain is primarily an artifact of the fundamental fluvial processes active within the floodplain through time (Blakey and Gubitosa, 1984; Mohrig et al., 2000; Tye, 2004; Peakall et al., 2007). Therefore, identifying these processes, which are relatively well understood, and the manner in which the processes operate within a specific floodplain can provide valuable information on how sediment is locally distributed over time (Hajek and Wolinsky, 2012).

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For most fluvial systems, stationary sediment that is exposed to fluvial flow is entrained dependent on [1] the relative force of the flow exerted on the sedimentary material and [2] the inertial force that controls the material's resistance to entrainment, which typically increases with the grain size and the cohesion of the material (Bagnold, 1956; Leopold et al., 1964; Dietrich and Smith, 1984). If the fluid force is greater than the inertial force, the sedimentary material is entrained by the flow and transported downstream. The balance of those two forces, in addition to the pre-existing, external material composing the environment containing them (e.g., the underlying rock in an incising system), controls much of the composition and organization of the river basin and its floodplain (Schumm, 1968). In an actively building (aggrading) floodplain within an alluvial basin, the river channel is composed of sediment deposited by flowing water within the channel banks. The associated floodplain is primarily composed of sediment transported and deposited by unchannelized water, typically by overbank floodwater. These two types of flow and their associated sedimentation regimes are responsible for the first-order organization of the architecture for most aggrading floodplains (Fielding, 1986; Paola, 2000). Sediment transport capacity and competency, which set the amount and maximum grain size of sediment transport, are exponentially related to the magnitude of force exerted by flow. Generally, unchannelized flow over the floodplain produces lower tractive forces than that occurring within the channel system. This phenomenon creates a spatial disparity in relative sediment grain size within the drainage basin, as fine-grained sediment becomes deposited upon the floodplain surface and relatively coarse-grained sediment becomes deposited within the channel network itself (Allan, 1974; Friend et al., 1979; Fielding, 1986). As the floodplain aggrades, it becomes filled with two significantly different types of deposits: fine-grained floodplain deposits composed of clays and silts and coarse-grained channel deposits composed of sands and gravels (assuming this range of grain sizes is present within the fluvial system). These two deposit types compose the alluvial architecture of interest for this study.

In this study, we examine the fundamental geomorphic processes responsible for the distribution of coarse and fine sediment within a floodplain. We employ a computational alluvial architecture model to quantify how different processes (e.g., lateral channel migration, floodplain sedimentation, avulsion) might alter floodplain development. By identifying how these processes affect the alluvial architecture of the floodplain, we propose that it becomes possible to predict characteristics of the alluvial architecture, including those with engineering significance. Alluvial architecture-type models have been successfully applied to identify the influence of geomorphic parameters and processes on the distribution of coarse-grained sediment deposits relative to finer-grained deposits, but few studies have attempted to use the modeling results outside of reservoir engineering and management applications (see North (1996), Bridge (2008), and Hajek and Wolinsky (2012) for eloquent reviews of alluvial architecture modeling). Alluvial architecture models explore how channel geometry, channel avulsion frequency, floodplain sedimentation rate, and uplift/subsidence affect the density of channel deposits within a floodplain cross section (Allen, 1979; Bridge and Leeder, 1979; Mackey and Bridge, 1995; Leeder et al., 1996; Gross and Small, 1998; Tornqvist and Bridge, 2002; Jerolmack and Paola, 2007; Willis and Tang, 2010). They also help define the amount of dependence that the geomorphic parameters typically have on each other (such as sedimentation rate and avulsion frequency) (Bryant et al., 1995; Heller and Paola, 1996).

The objectives of this study are: [1] to examine how the magnitudes of specific geomorphic processes affect the alluvial architecture of a floodplain cross section with focus on the distribution of channel deposits (i.e.,sand) within the cross section, and [2] to explore how the results of a computational alluvial architecture model can aid a geotechnical investigation for an engineering project.

We initially calibrate our model with data collected from a geological investigation in support of a levee-engineering project and validate our model results against the investigation's observations. We employ an additional range of model input parameters, beyond that observed in the project area, to better identify how fluctuation of these parameters can affect the dimensions and distributions of the channel deposits within modeled floodplain cross sections.

2. Study site

This study employs hydrologic, geologic, and topographic data collected from the upper Trinity River drainage basin near Dallas, Texas (Figure 1). These data are used to calibrate the model boundary conditions and initial parameters, as well as validate the model output. The Trinity River floodplain near Dallas is the site of multiple civil-works levee construction and rehabilitation projects, some of which date back to the early 20th century (Roig-Silva et al., 2010). The types and densities of geologic data collected for these projects illustrate the typical data demand, availability, and collection procedures common to modern infrastructure engineering projects.

The Trinity River at Dallas drains an area of approximately 15,800 km^2. The city of Dallas lies immediately downstream of the confluence of the Elm Fork and the West Fork of the Trinity River system. Three distinct quaternary terraces have been identified surrounding the modern floodplain. The lowest and youngest of these terraces (formed 30 to 76 ka BP and referred to as the Hickory Creek Terrace) lies approximately 20 to 30 m above the modern floodplain at its location near Dallas (Ferring, 1990). The north and south faces of the lowest terrace are well matched in this area and range from being approximately 6-km apart at the upstream reach of the study area to being 1.5-km apart at the downstream extent of the study area. Municipal surface-water reservoirs were installed on both upstream tributaries of the Trinity River in the 1960s. Pre-1960 discharges for the 2-, 10-, and 50-year floods were 602, 1926, and 3943 $m^3\ s^{-1}$, respectively. Post-1960 discharges for the 2-, 10,- and 50-year floods have been reduced to 594, 1337, 2105 $m^3\ s^{-1}$, respectively.

The modern Trinity River levee system was built in response to the initiation of the Dallas Floodway project in the 1920s. The floodway project led to the eventual realignment of the Trinity River channel system to a parallel route approximately south of its natural course. A length of 36.4 km of 10-m tall levees surrounds the current channel system, constraining it to a 0.5–1.0-km wide floodplain. Before levee systems were in place, anecdotal evidence described large floods that inundated the full basin floor between the Hickory Creek terraces (Tompkins et al., 2010).

Geologic and geotechnical investigations carried out in support of the various Dallas Floodway projects have collected and synthesized more than 2000 geologic borings for the project area. However, the vast majority of the borings are limited to the area immediately surrounding the levee system (which approximates one-third to one-half of the modern floodplain width) and many borings are not deep enough to reach bedrock. Additional boring data from highway and bridge locations were analyzed for this study to better characterize the full basin width, which were provided by the Texas Department of Transportation. Boring data indicate that the bedrock (Cretaceous shale, chalk, and sandstone from the Eagle Ford and Austin Chalk Formations) that underlies the modern floodplain sediments typically lies between 5 to 20 m below the floodplain surface, dipping to the downstream (southeast) direction. Aerial imagery and topographic maps for the period preceding the channel realignment were collected and geo-referenced to identify the natural river channel planform. Within the study area, the modern Trinity River appeared to be of moderate sinuosity (1.5 to 1.7) with channel widths of 50 to 70 m and average meander amplitude near 800 m.

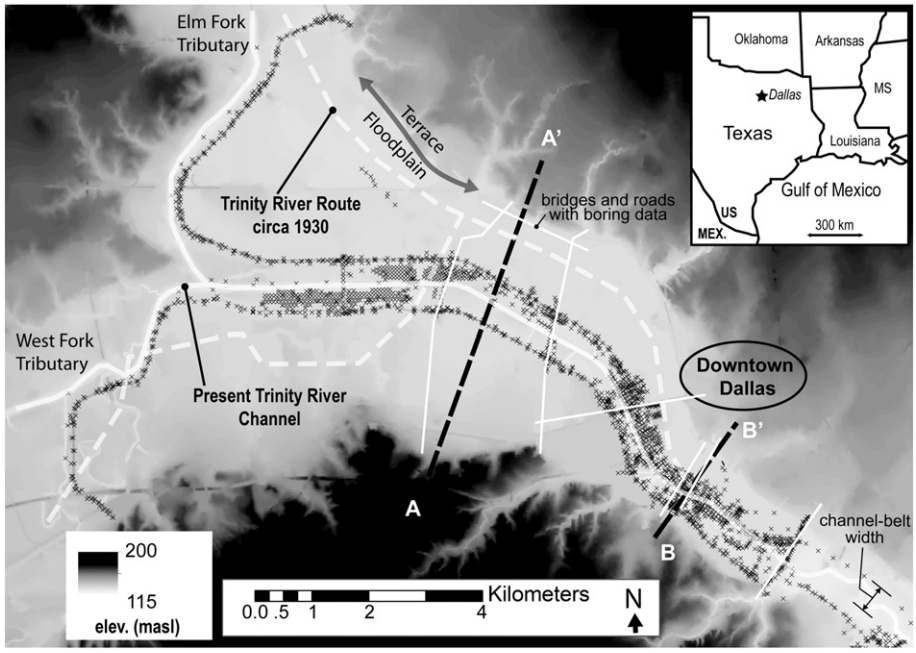

Fig. 1. Map of the Trinity River basin near Dallas, Texas. The bimodal distributions in topographic elevation differentiate the modern floodplain and the surrounding terraces. The river flows to the southeast. This map illustrates the modern (thick solid white line) and pre-realignment (dashed white line) courses of the Trinity River. The small black crosses show the location of the geologic boring data collected in support of the levee projects. The thin solid white lines show the location of roads and bridges with associated boring data used by this study.

3. Methods

3.1. Model description

This study employs a two-dimensional, computational alluvial architecture model which computes synthetic stratigraphy of a floodplain cross section within an alluvial basin. The model builds a floodplain cross section based on process-imitating rules derived from fluvial geomorphology and sedimentology that control the processes of floodplain sedimentation rate, channel avulsion, and sediment compaction (i.e., consolidation). The computed stratigraphy differentiates between floodplain sediment deposited by channel processes (assumed to be sand) and sediment deposited by overbank flooding (assumed to be clay).

The model employed by this study is based on the computational model described in detail by Mackey and Bridge (1992) and the improvements to the model algorithms, which relate to avulsion prediction and the overbank aggradation rate, described by Mackey and Bridge (1995). Previous applications of the Mackey and Bridge (1992, 1995) model are presented in Heller and Paola (1996), Leeder et al. (1996), Bridge (1999), Bridge and Tye (2000), Karssenberg et al. (2001), and Tornqvist and Bridge (2002). This type of model is referred to as a "LAB" alluvial architecture model after the researchers of the underlying theory, Mike Leeder, John R.L. Allen, and John Bridge (e.g., Allan, 1974, 1978; Bridge and Leeder, 1979).

Typically, LAB models fix the dimensions of a channel deposit to static mean values approximate to the average channel-belt width and the bankfull channel depth. In sedimentology research, channel-belt width is often estimated as the maximum amplitude of a meander loop (Allan, 1974). The model employed in this study initially sets the channel deposit width to one bankfull channel width but allows it to increase with time based on a defined lateral (i.e., cross-basin) migration rate(s). For this study, lateral migration rates were calibrated using bank erosion rates measured along the modern river. The lateral migration mimics a meander sequence (i.e., the formation of a meander loop with the assumption that the loop is bisected by the modeled cross section). The direction of the migration is randomly selected and will continue until one of the two following events occurs: a probability-based sequence termination or a process-based avulsion.

The probability that a meander sequence will terminate at any one time was calibrated to reproduce the natural sinuosity of the Trinity River. The termination can occur as a neck cutoff, chute cutoff, or a probability-based avulsion. Typically, neck cutoffs do not occur until a meander loop approaches its maximum probable amplitude (Howard, 1996; Micheli and Larsen, 2011). Probability-based avulsions simulate avulsions that occur upstream of the modeled cross section and are not influenced by processes occurring at the modeled cross section. If a cutoff occurs, a new meander sequence will initiate. If an avulsion occurs, the channel is relocated and inset into the nearest section of floodplain at the minimum surface elevation.

A process-based avulsion requires two conditions: the river must be in flood stage and the channel bed elevation must be higher than the minimum elevation of the local floodplain. The difference between the channel bed and floodplain elevations is referred to as the channel 'superelevation'. If the two conditions are met, the probability of an avulsion will be triggered at that time based on the current magnitude of superelevation (Figure 2). Interpretation of observed geologic stratigraphy suggests that avulsion frequency increases rapidly as the elevation of the channel bed becomes perched above the surrounding floodplain surface elevation (Mohrig et al., 2000; Jerolmack and Mohrig, 2007); this increase in avulsion frequency results in the superelevation seldom exceeding two or three channel depths in magnitude.

The model computes a flood regime for each modeled time interval. The regime values are sampled randomly from a synthetic probability distribution calibrated by observed Trinity River discharge data. The maximum flood discharge for each time interval is used to estimate the probability of an avulsion or channel cutoff and the lateral extent of overbank sediment deposition due to flooding during that time interval. The number of peak annual floods that exceed the bankfull discharge during each time interval is used to calculate the thickness of channel and overbank sediment deposition for that interval.

Fig. 3 shows a simple example of how the model builds a floodplain cross section within the computational grid. Grid dimensions are set to scale with the dimensions of the basin in which the

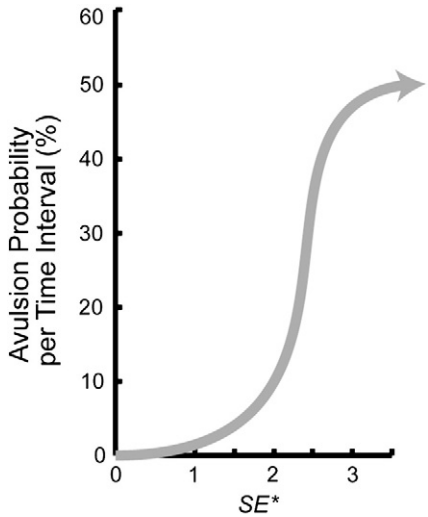

Fig. 2. The probability, as employed by this study, that a 'process-based' avulsion will occur versus SE^* per time interval; $SE^* =$ channel superelevation/bankfull channel depth.

floodplain is built. The grid cells comprising the initial floodplain in Fig. 3A are primary composed of fine sediment (clay), except for the cells comprising the initial active channel position, which are composed of coarse sediment (sand). Fig. 3B shows the grid after time has elapsed (i.e., $t_i = [Time\ at\ B] - [Time\ at\ A]$). In this diagram, the channel has aggraded and laterally migrated. The dimensions of the channel deposit now include the integration of the area of the cells that were previously occupied by the channel. The floodplain has aggraded based on predefined geomorphic rules that set the sedimentation rate within a given floodplain cell as a function of the channel aggradation rate, the computed flood regime, and the cell distance from the active channel. Fig. 3C illustrates how superelevation is computed for the computed scenario and where the probable new active channel location would be if an avulsion was triggered. Fig. 4 shows what the hypothetical plan view of the simulated river and channel deposits in Fig. 3A and B might look like. Fig. 5 summarizes the full model algorithm employed by the alluvial architecture model.

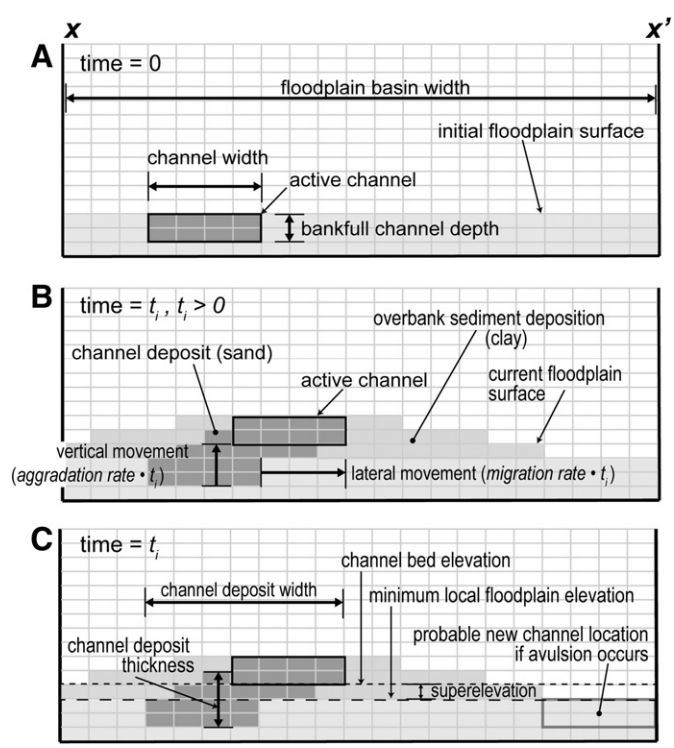

Fig. 3. A conceptual diagram illustrating how the model constructs a floodplain within a computational grid.

3.2. Description of the experiments

Two floodplain cross sections were modeled, A – A' (i.e., X-sec A) and B – B' (i.e., X-sec B). The cross section locations are shown in Fig. 1. For both cross sections, the model's computational grid was set to produce a horizontal resolution of 10 m and a vertical resolution of 0.01 m. Five different scenarios were modeled at each cross section, producing ten different experiments (Table 1). The 'channel aggradation per annual flood' value found in Table 1 is the increase in channel elevation due to the deposition of channel sediment during the annual peak flow period for each year of the modeled time period, if the annual peak flow is predicted to exceed the channel bankfull flow capacity. The data used to calibrate the model to the Trinity River basin near Dallas, TX are shown in Table 2. The 'base case' scenarios, Experiments 1 and 6, are the best fit to the calibration data. The 'dynamic climate' scenarios, Experiments 2 and 7, used time-dependent values for the channel dimensions and channel aggradation rate to simulate the effects of the paleoclimate reported in Ferring (1990) (Figure 6). The 'double sedimentation rate' scenarios, Experiments 3 and 8, were similar to the base cases except that the channel aggradation rate was doubled. The 'complex initial topography' scenarios, Experiments 4 and 9, were conducted using a non-uniform base level to simulate the effect of a more diverse initial topography at the beginning of each model run. This topography was estimated from boring data, which suggested that, initially, the channel was incised within a bedrock canyon approximately 10 m below the mean elevation of the contemporaneous floodplain (Roig-Silva et al., 2010). Experiments 5 and 10 were run by combining the dynamic climate and the complex initial topography scenarios. The different scenarios were designed to test how different geomorphic-related conditions affect the model output.

The model terminated after the conclusion of a meander sequence once the total elapsed time exceeded 11,000 years; it is assumed that the floodplain sediments have been deposited since the last sea-level low stand at the beginning of the Holocene. The modern (pre-realignment) lateral position of the channel along each floodplain cross section was selected to be the initial channel location for each model run. This ensures that the model will predict at least one channel deposit at the last known channel location before the engineered realignment.

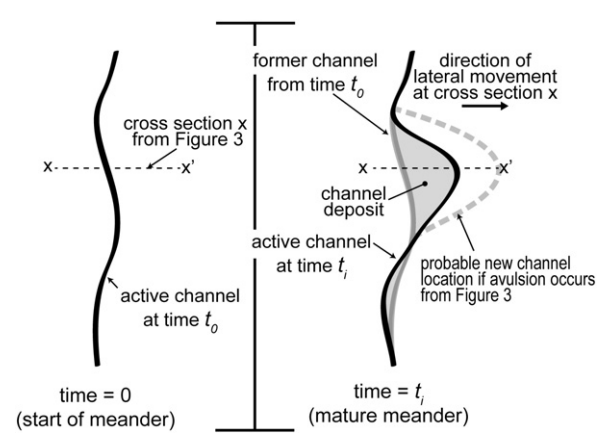

Fig. 4. Plan view of the river channel conceptualized in Fig. 3.

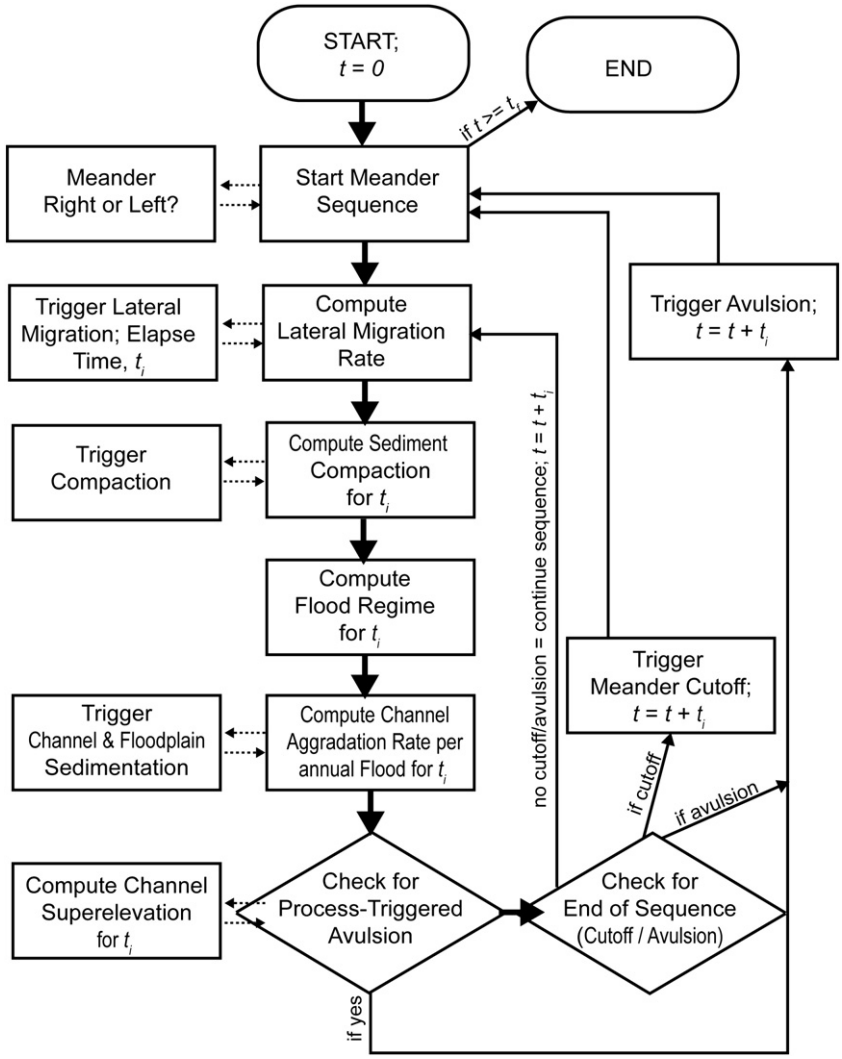

Fig. 5. Diagram of the alluvial architecture model algorithm. For this study, the modeled time period (t_f) equals 11,000 years.

The alluvial architecture model was run one hundred times (simulating the construction of one hundred floodplain cross sections) for each of the ten experiments. Sensitivity analysis found that one hundred iterations were adequate to identify the average tendencies produced by the model.

3.3. Channel deposit measurement

After each model run, the computed dimensions of every individual channel deposit were recorded. The deposit width was computed as the maximum lateral extent of the entire deposit body (Figure 7). The mean

Table 1
Summary variables for the Experiments 1–10.

	Experiment									
	1	2	3	4	5	6	7	8	9	10
Scenario	*Base case*	*Dynamic climate (DC)*	*Double Sedimentation Rate*	*Complex initial topography (CT)*	*DC + CT*	*Base case*	*Dynamic climate (DC)*	*Double sedimentation rate*	*Complex initial topography (CT)*	*DC + CT*
Cross section	A	A	A	A	A	B	B	B	B	B
Channel width: depth mode value	20:1	20:1 (dry periods), 30:1 (wet periods).	20:1	20:1	20:1 (dry periods), 30:1 (wet periods).	20:1	20:1 (dry periods), 30:1 (wet periods).	20:1	20:1	20:1 (dry periods), 30:1 (wet periods).
Channel-belt width mode value (m)	800	800 (dry periods), 400 (wet periods).	800	800	800 (dry periods), 400 (wet periods).	800	800 (dry periods), 400 (wet periods).	800	800	800 (dry periods), 400 (wet periods).
Channel aggradation per annual flood (mm)	4	2–5 (4 avg.)	8	4	2–5 (4 avg.)	3	1.5–4 (3 avg.)	6	3	1.5–4 (3 avg.)

Table 2
Input values for the primary model calibration parameters.

Variable	Units	X-sec A	X-sec B	Comments
Basin width	m	4500	1500	Width of the drainage basin in which the floodplain is built; based on measurement of the modern Trinity River.
Avg. channel-belt width (CBW_{Mean})	m	800	800	Width of floodplain affected by active channel; scales with meander amplitude. Based on measurement of the modern Trinity River.
Bankfull channel width	m	60	60	Based on measurement of the modern Trinity River.
Bankfull channel depth	m	3	3	Based on measurement of the modern Trinity River.
Avg. floodplain aggradation rate	m year^{-1}	1.1×10^{-3}	1.4×10^{-3}	Avg. depth of floodplain sediments divided by the time of deposition; interpreted from boring data reported in Roig-Silva et al. (2010).
Floodplain sediment compaction rate	m year^{-1}	1.0×10^{-4}–5.0×10^{-4}	1.0×10^{-4}–5.0×10^{-4}	Dependent on the overall thickness of overbank sediment and overburden thickness; based on typical rates for Holocene sediments described in Reed and Yuill (2009).
Bank erosion rate	m year^{-1}	0.33–1.0	0.33–1.0	Controls the channel lateral migration rate. Varies in time based on computed CBW_i/CBW_{Mean}; rates are based on observed values reported in Tompkins et al. (2010).
Bankfull discharge	m^3 s^{-1}	425	425	Estimated from USGS gauge data (at Rosser, TX) as the 1.5-year flood.
Avg. annual flood (F_{Mean})	m^3 s^{-1}	963	963	Average value of the peak annual discharge; calculated from USGS gauge data (at Rosser, TX).
Estimated flood width	m	$(F_i/F_{Mean}) \times 500$	$(F_i/F_{Mean}) \times 500$	Width of floodplain inundated by the F_i flood; interpreted from observations in Tompkins et al. (2010).
Probability-based avulsion frequency	% annually	0.1	0.1	Estimated reoccurrence interval for a channel avulsion; based on analysis of East Texas river channels Phillips (2007).

CBW_i = channel-belt width at time interval t_i (i.e., current time of calculation); F_i = computed annual flood for time interval t_i.

thickness of a channel deposit was computed by (1) adding up the total depth of the channel deposit sediment per column of the computational grid area that composed the deposit and (2) calculating the average of those values. The maximum deposit thickness equals the largest computed depth of channel deposit sediment per column of the computational grid for each deposit.

At the conclusion of each experiment, the distributions of each deposit dimension computed during the one hundred model runs were recorded and summary statistics (i.e., average, standard deviation) of the distribution values were calculated.

4. Results

4.1. Computed deposit abundances and dimensions — base case scenarios

Fig. 8 shows an annotated example of a floodplain cross section at the conclusion of one model run from Experiment 1. During the illustrated example, five separate channel deposits were formed. Three of these channel deposits experienced some spatial overlap. Approximately 3.5 km of the 4.5-km floodplain width contained sand from a channel deposit. Channel deposits ranged from 1 to 3 bankfull channel depths (3 to 9 m) in thickness and from 300 to 1200 m in width. Note that the combination of simultaneous lateral migration and vertical channel aggradation produce channel-deposit shapes that slope upward in the direction of the channel migration. This phenomenon is observed in other studies predicting meandering-channel geometry (i.e., Bridge, 1975; Gross and Small, 1998) and is visually accentuated by the vertical exaggeration employed to illustrate a modeled cross section.

The results from Experiment 1 represent the base case for X-sec A. For the base case, there was an average of 4.2 channel deposits per modeled cross section with an average deposit width of 837 m (Table 3). The averaged mean thickness of a channel deposit was 5.3 m and the averaged maximum thickness per channel deposit was 7.7 m. The thickness of a channel deposit predominately depended on the frequency in which the channel reoccupied the same section of floodplain (e.g., due to meander loop cutoffs, which reset the channel to near its location at the initiation of a meander loop) during an inter-avulsion period. Experiment 6 represents the base case for X-sec B. For the two base cases, X-sec B contained a smaller average number of channel deposits than X-sec A (3.0); however, the X-sec B deposits were, on average, wider (919 m) and thicker (7.0 m).

The accuracy of the computed alluvial architecture for the base case scenarios (i.e., Experiments 1 and 6) was assessed by comparing the model results to the architecture interpreted from geologic boring data (Table 4). Fig. 9 shows two geologic cross sections, which were interpreted from the available boring data for the same approximate

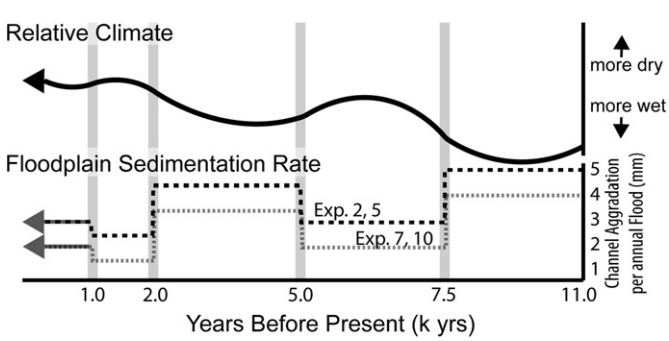

Fig. 6. The relative climatology and sedimentation regime employed in the "dynamic climate" scenario. The climate oscillates but becomes drier in time. Channel aggradation is set proportional to relative climate wetness, therefore, the floodplain sedimentation rate generally decreases in time. The sedimentation rate is fit to the general regime described in Ferring (1990) for the upper Trinity River drainage basin.

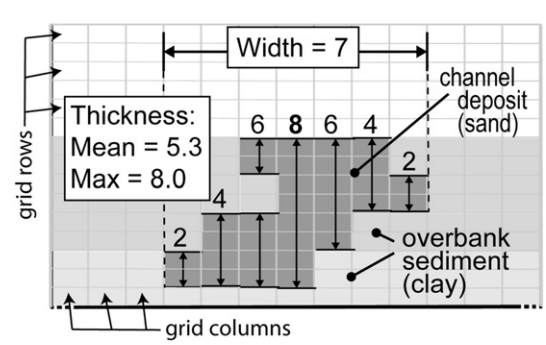

Fig. 7. Example of how width, mean thickness, and maximum thickness are calculated for a channel deposit simulated within the computational grid. The mean thickness equals the sum of the sand thickness for each grid column divided by the number of grid columns occupied.

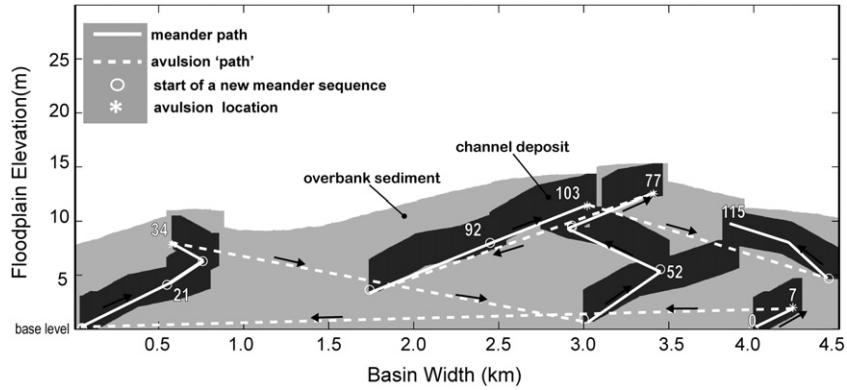

Fig. 8. Example of a floodplain cross section computed for the base case of X-sec A (i.e., Experiment 1). The white lines indicate the evolving position of the channel within the floodplain during the floodplain development. The numbers indicate the elapsed model time when the channel was at that position (in 100-year increments).

locations as the modeled cross sections, X-sec A (9.A) and X-sec B (9.B). Assuming that [1] the sand facies identified in neighboring borings are both parts of the same deposit and that [2] the sand deposits were formed by channels oriented parallel to the longitudinal down-basin slope (NW to the SE), there appears to be three individual channel deposits per cross section. This might indicate that at least two avulsions took place during floodplain construction. These channel deposits have a similar mean thickness near 5 m, which is approximate to the mean thickness values predicted during the model experiments. The deposit widths appear larger in the cross section from Fig. 9A than in the cross section from Fig. 9B, which is in contrast to the values predicted by the computational model.

The fraction of the floodplain widths underlain with sand (≈70%) is approximately equal for both interpreted cross sections. In contrast, the model computed that the floodplain near X-sec B would be almost completely underlain by sand and that the channel deposit widths would be, on average, 214% smaller than the interpreted values. The magnitude of the discrepancy in channel deposit characteristics between the interpreted and modeled cross sections near X-sec B may suggest errors in model calibration relating to the two-dimensionality of the model. The model does not explicitly simulate upstream or downstream effects that impact the alluvial architecture of the cross section. For example, the model predicted approximately twice the frequency of avulsions in the cross section in X-sec A than in X-sec B. However, because X-sec B is downstream of X-sec A, an avulsion occurring in X-sec A would likely impact the channel location in X-sec B too. Underestimating the amount of avulsions that would occur upstream of a simulated cross section would, in general, lead to an underestimation of the amount of channel deposits in that cross section. This could account for the fact the model under-predicted the amount of channel deposits in X-sec B relative to the interpreted amount in Fig. 9. Future modeling efforts could use model results from upstream cross sections to better inform calibration values (e.g., upstream avulsion probabilities) for cross sections modeled in the downstream direction.

The interpreted cross section in Fig. 9A does show channel deposits spatially distributed in the approximate pattern predicted by the model (i.e., located at the floodplain margins and in one area within the middle of the cross section). Note that if the real-world alluvial architecture did not conform to the two assumptions stated above, the channel deposit widths interpreted from geological boring data would likely be over-predicted by the interpretation methodology.

A report by Roig-Silva et al. (2010) contains geologic data for fifteen additional cross sections within the Trinity River levee project area. However, these cross sections are restricted to the modern leveed floodplain and, therefore, the cross sections only span a fraction of the full basin width. The dimensions of nine channel deposits can be identified from these partial cross sections and are summarized in Table 4.

4.2. Computed deposit abundances and dimensions — alternative scenarios

The different alternative experimental scenarios produced significant variation within the distributions of computed channel deposit dimensions. Fig. 10 displays the distribution of channel deposit widths computed during each experiment. The summary statistics for the distributions of the channel deposit dimensions are listed in Table 3.

Table 3
Summary statistics for the deposit dimensions modeled in Experiments 1–10.

Exp. #	Deposits per X-sec	Deposit dimensions					
		Width (m)		*Mean thickness (m)*		*Max thickness (m)*	
	Avg.	*Avg.*	*Stdev.*	*Avg.*	*Stdev.*	*Avg.*	*Stdev*
1	4.2	837	334	5.3	1.4	7.7	2.3
2	4.9	772	325	5.1	1.4	7.3	2.3
3	6.8	649	234	5.2	1.3	7.7	2.2
4	4.3	796	288	5.4	1.8	7.8	3.1
5	4.9	699	261	5.8	2.3	7.8	3.2
6	3.0	919	337	7.0	2.0	9.5	3.1
7	3.5	842	324	6.6	2.0	8.8	3.1
8	4.3	793	316	7.0	2.2	10.1	3.5
9	3.3	808	336	6.5	2.3	9.1	3.6
10	3.6	757	317	6.6	2.3	8.7	3.3

Table 4
Comparison of the channel deposit dimensions interpreted from geologic data vs. those computed during the experiments.

	Mean thickness (m)	Max thickness (m)	Width (m)
X-sec A			
Interpreted Deposit 1	5.0	7.0	700
Interpreted Deposit 2	5.3	6.5	1800
Interpreted Deposit 3	4.0	5.5	400
Avg. interpreted values	4.8	6.3	967
Avg. computed values	5.3	7.7	837
% Difference, interp. vs. comp.	−11%	−22%	+13%
X-sec B			
Interpreted deposit 1	5.0	6.5	200
Interpreted deposit 2	6.0	7.0	175
Interpreted deposit 3	6.0	7.8	500
Avg. interpreted values	5.7	7.1	292
Avg. computed values	7.0	9.5	929
% Difference, interp. vs. comp.	−23%	−33%	−214%
From Roig-Silva et al. (2010) Avg. of nine partial X-secs	7.0	9.0	650

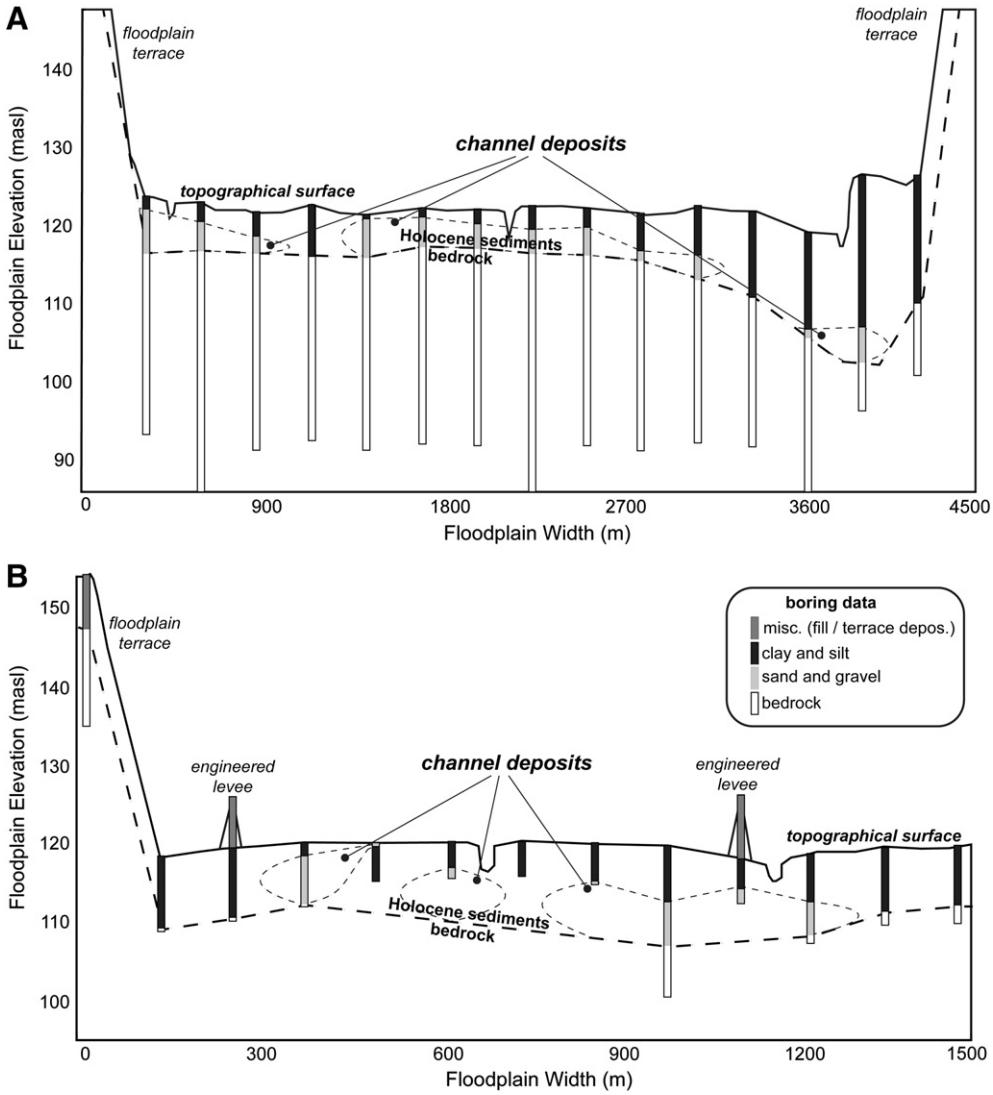

Fig. 9. The alluvial architecture for the floodplain cross sections approximate to (A.)X-sec A and (B.)X-sec B as interpreted from available geologic boring data.

The channel deposit widths computed in X-sec B were limited in size to less than 1500 m (i.e., the basin width). However, the experimental scenarios run in X-sec B typically produced larger channel deposits widths than the scenarios run in X-sec A. Channel deposit width values did not display a robust correlation with channel deposit thickness values. Fig. 11 shows the relationships between width and thickness for individual channel deposits computed during Experiments 1, 4, and 6.

The alternative scenarios, represented in Experiments 2 through 5 (X-sec A) and 7 through 10 (X-sec B), produced some fundamental differences in the floodplain architecture as compared to that produced by the base case scenarios. Fig. 12 shows cross sections that were computed from four of the alternative scenarios and illustrates these differences. The alternative scenarios typically increased the number of channel deposits and decreased the channel deposits widths as compared to the base case scenarios. Channel deposit thicknesses were marginally decreased during the double sedimentation rate scenarios relative to the base case scenarios. Increasing the base-level complexity increased the channel deposit thickness in X-sec A and decreased it in X-sec B.

4.3. Avulsion frequency

Table 5 shows the frequency of process-based avulsions for each of the ten experiments and compares those frequencies to the average frequency value computed for X-sec A and X-sec B. The average process-based avulsion frequency for X-sec A was greater than twice that of X-sec B (2.8 avulsions per model run vs. 1.2 avulsions per model run). Experiments 3 and 8, which included doubling the steady-state sedimentation rate, experienced a significantly higher avulsion frequency than the other experiments. Fig. 13 shows that process-based avulsion frequency computed for each experiment trended with both the average number of channel deposits per cross section (positive) and the average channel deposit dimensions (negative).

4.4. Fraction of the floodplain width with underlying channel deposits

Fig. 14 displays the fraction of the floodplain width computed to contain channel deposits (i.e., sand) for the one hundred model runs of each experiment. For the vast majority of model runs, 50 to 85% of the X-sec A width contained sand and 90 to 99% of the X-sec B width contained sand. Fig. 15A and B displays the average value of the sand thickness in each column of the model's computational grid at the conclusion of the individual model runs for the base case scenarios. The sand thickness per grid column is the total thickness of channel deposit sediment (i.e., sand) for each grid column. The average sand thickness for each grid column is the average

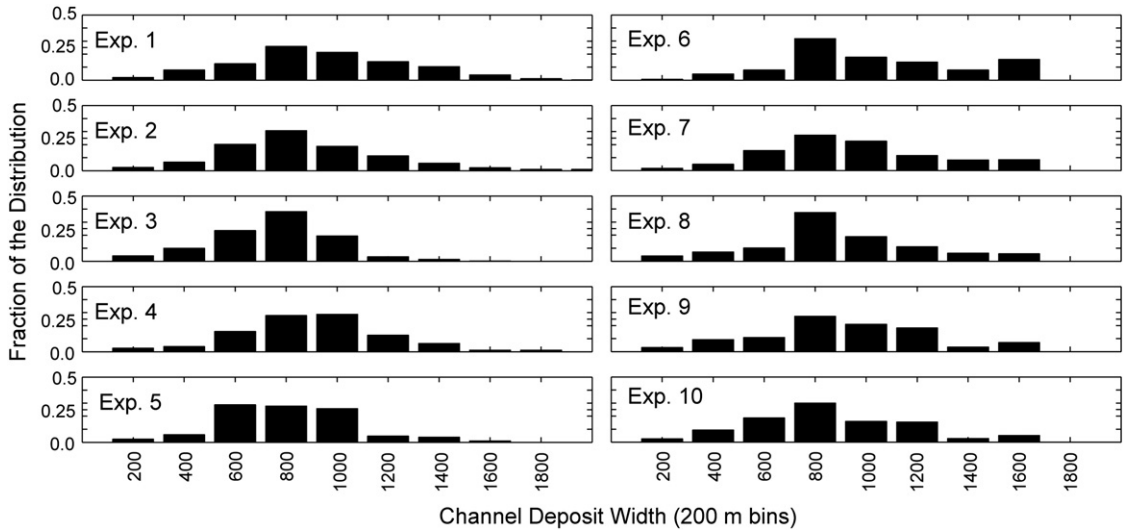

Fig. 10. Histogram of the total distribution of channel deposit widths computed during the 100-model runs for each experiment.

value after a typical model run. The average sand per grid column scales with the frequency that the unit of floodplain width was occupied by the active river channel during a typical model run. For Experiment 1, the average value of the total sand thickness ranged from 1 m at the middle of the cross section to 5 m at the cross-section margins. For Experiment 6, the total sand thickness per unit width was fairly uniform along the entire cross section and averaged 8 m.

Fig. 15C and D shows the average value of the sand width in each row of the model's computational grid at the conclusion of the individual model runs for the base case scenarios. The sand width per grid row is the total horizontal length of channel deposit sediment in each grid row. The average sand width per grid row scales with the frequency that the unit of floodplain depth was occupied by the active channel. Fig. 15C indicates that, for Experiment 1, the prevalence of subsurface sand within the floodplain peaks at an elevation approximately 8 m above the initial base level and drastically decreases with increasing or decreasing depth relative to that elevation. Fig. 15D indicates that, for Experiment 6, the prevalence of subsurface sand peaks at an elevation approximately 5 m above the initial base level and remains relatively stable until an elevation exceeding 15 m above the initial base level. Fig. 16 is similar to Fig. 15, except that it displays data for the experiments with complex initial topography scenarios (i.e., Experiments 4 and 9). For these experiments, the spatial distributions of the total sand thicknesses were more variable than the base case scenarios. For Experiment 4, the average value of the total sand per grid column typically ranged between 0 and 10 m and, for Experiment 9, the average value of the total sand per grid column ranged between 5 and 14 m. The segment of the floodplain widths that contained the incised channels set into the initial base level typically held the largest sand thicknesses. Average sand width per unit floodplain depth increased as the floodplain width at each depth interval increased.

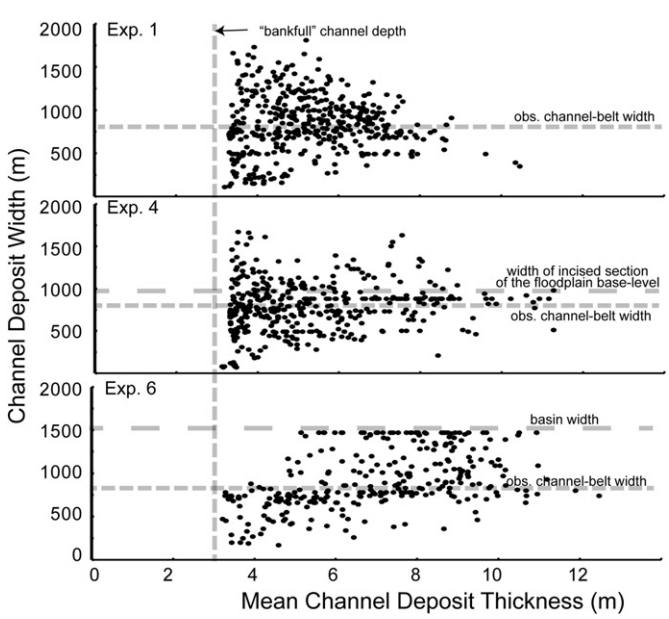

Fig. 11. The relationship between channel deposit thickness and width for Experiments 1, 4, and 6. The widths of the observed channel-belt width (800 m), the incised bedrock channel in Experiment 4, and the total basin width in Experiment 6 are shown for reference.

5. Discussion

5.1. Interpretation of the results

5.1.1. Deposit dimensions

The base case experimental scenarios (i.e., Experiments 1 and 6) produced the widest channel deposits on average. For the other scenarios, the main factors that appeared to constrain the channel deposit widths were the channel aggradation rate and the initial base-level topography. The influence of the channel aggradation rate was due to its positive impact on channel superelevation. In the employed model, superelevation was the primary trigger for process-based avulsions. Avulsions had the net effect of spatially disconnecting the locations of channel sediment deposition, ending the growth period of one channel deposit at one location and starting the growth period of another deposit at a new distal location. Experiments 3 and 8, which doubled the channel aggradation rate relative to the base case scenarios, had the highest avulsion frequencies, which likely led to the fact that they also had the largest number of channel deposits per cross section and the smallest individual deposit widths.

The scenarios that employed a complex initial topography simulated floodplain aggradation on a base-level surface with an incised channel. The reduced basin width within the incised segment of the floodplain increased the probability that a basin wall would impede the modeled lateral migration of the channel. Limiting the lateral extent of the channel migration would also limit the lateral extent

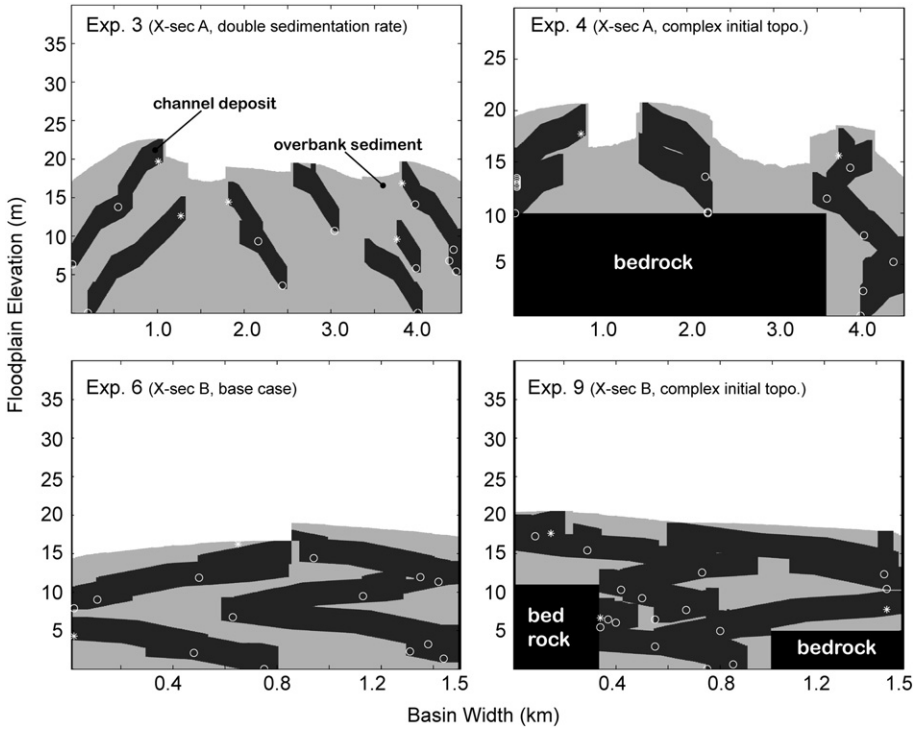

Fig. 12. Example floodplain cross sections computed during 4 different experiments. The figure uses the same symbology as that described in Fig. 8. The plots for Experiments 4 and 9 display the geometry of the incised channels modeled by the complex initial topography scenarios.

of the channel deposits. On average, this limitation would lead to smaller deposit widths within the computed architecture of the floodplains with incised channels than within the non-incised, and therefore wider, floodplains. For example, while the floodplain was aggrading within the incised channel in Experiments 4 and 5, the effective floodplain width was 1 km; therefore, the maximum possible deposit width during this period was 1 km.

Experiments conducted in X-sec B typically produced both fewer and larger channel deposits than did experiments conducted in X-sec A. While the smaller floodplain width in X-sec B limited the maximum width values to values smaller than that observed in X-sec A (see Figure 11), experiments in X-sec B routinely produced a higher percentage of very wide deposits (i.e., greater than 1000 m) (see Figure 10). This is likely because the model predicted more uniform flooding across the relatively small floodplain in X-Sec B than in X-sec A (i.e., a flood inundated a larger fraction of the floodplain width in X-sec B than in X-sec A). More uniform flooding produced more uniform flood-dependent overbank sediment deposition across the floodplain width. For experiments in X-sec B, the reduced lateral gradient in overbank deposition produced less topographic relief and reduced the rate of channel superelevation relative to experiments in X-sec A. This effect was enhanced by the reduced channel aggradation rates employed during experiments in X-sec B relative to the channel aggradation rates employed during experiments in X-sec A (e.g., 3 mm per annual flood versus 4 mm per annual flood for the base case scenarios). The average effect of these differences resulted in a reduction of the process-based avulsion frequency by 50% relative to that observed in X-sec A.

Experiments in X-sec B produced higher deposit thicknesses than experiments in X-Sec A. This observation is likely correlated to the relative longevity of the growth period for these deposits due to the less frequent avulsions.

The results of the experiments show no significant relationship between the mean or maximum channel deposit thickness and the deposit width. A relationship between deposit thickness and width would be advantageous to engineers because the deposit-thickness data observed in boring logs could be used to extrapolate the associated deposit width. Fig. 11 shows that deposit widths tend to cluster at distances influenced by the modeled boundary conditions, such as the observed channel-belt width (i.e., 800 m, which was used to determine

Table 5
The average and relative number of process-based avulsions per cross-section.

Exp. #	Avulsions per X-sec (AV_i)	AV_i/AV_{AVG}
1	2.3	0.8
2	2.5	0.9
3	5.0	1.8
4	2.2	0.8
5	2.2	0.8
Average for Exp. 1–5 (AV_{AVG}) = 2.8		
6	0.8	0.9
7	1.1	1.2
8	2.1	2.4
9	0.9	1.0
10	1.0	1.1
average for Exp. 6–10 (AV_{AVG}) = 1.2		

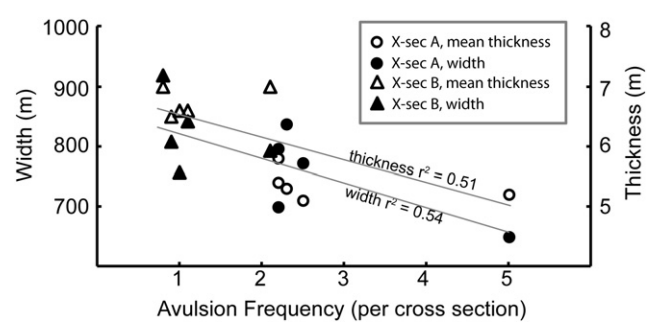

Fig. 13. The relationship between the avulsion frequency and averaged deposit dimensions (i.e., mean thickness and width) for the 10 experiments.

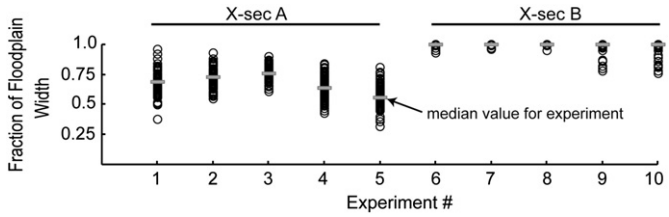

Fig. 14. Fraction of the floodplain width with sand. The complete distribution of the computed values (open black circles) and the median value (gray dash) for each experiment are shown.

the probability that a meander cutoff or a probability-based avulsion would occur) and floodplain widths (which limits the extent to which a deposit can grow). Deposit thicknesses showed little clustering at particular elevations above the minimum value (i.e., bankfull channel depth).

5.1.2. Deposit distribution

Theoretically, the fraction of a floodplain width that contains sand will increase, on average, due to increases in [1] the width of the channel deposits and [2] the total number of channel deposits within the floodplain cross section. However, the model data show that avulsion frequency, a primary control of these two influences, affects each influence in opposing ways. Increasing the avulsion frequency reduces the average channel deposit width but increases the overall number of channel deposits. This phenomenon likely reduces the relative effect of each scenario on the predicted fraction of the floodplain that contained sand during each experiment.

For the experiments conducted in X-sec A, the fraction of the floodplain width without any sand underneath it ranged from 15 to 50% on average, dependent on the modeled scenario. Experiments 1, 2, and 3 produced floodplain widths containing segments without any sand. However, the locations of these segments were not spatially consistent for every floodplain cross section simulated during the experiments. For the experiments conducted in X-sec A, the sand was consistently thinnest in two segments located at 1.0 to 1.5 km and 3.0 to 3.5 km along the floodplain width (measured using the south edge of the floodplain as the origin). Experiments 4 and 5 produced very thin sand thicknesses in these segments, including smaller subsegments that typically contained no sand (Figure 16). The locations of the floodplain segments that routinely had abnormally high or low amounts of sand appear to be from compensatory channel-stacking processes inherent within the modeled processes. Compensatory stacking results from the tendency for channels to change from positions of relatively high relief to positions of relatively low relief, if given an opportunity. During avulsions, areas remote from channel activity that have retained relatively low elevations due to low rates of overbank sediment deposition are more likely to become new locations of the channel than areas close to the channel that were frequently flooded. Evidence of compensatory stacking is common in certain sedimentary systems (Straub et al., 2009; Hajek et al., 2010; Hofmann et al., 2011). In this model, the location of the topographic low is set by [1] the channel location, [2] the lateral distance away from the active channel in which floodplain sediment can be readily deposited during floods, and [3] the total basin width. For the floodplain sedimentation regimes implemented within this model, the proximal topographic low (i.e., the closest lateral location at the lowest floodplain elevation) typically never was closer than 2 km from the active channel position. This promoted the clustering of channel deposits, first at the initial channel position at the start of each model run, and then extending laterally away at approximate 2-km intervals. Past research on river avulsions in fluvial systems has identified additional geomorphic factors that influence channel deposit stacking in other ways, such as the tendency of antecedent topography and gradients in soil erodibility to promote the reoccupation of paleochannels in time (e.g., Aslan et al., 2005; Jerolmack and Paola, 2007). However, because of the simplicity of the model employed in this study, these factors were explicitly not considered.

Channel deposits did not appear clustered in X-sec B, where the total sand thickness was more evenly distributed along the floodplain width relative to X-sec A (see Figure 15B). The more uniform lateral deposition of floodplain sediments during floods in X-sec B prevented the build-up of large lateral topographic gradients, which suppressed the activation of channel-stacking patterns within the sedimentary system.

5.2. Comparison with channel deposits observed in nature

Fig. 17 illustrates the distribution of modeled width-to-thickness (w/t) channel deposit dimensions computed in this study in relation to observed values reported in literature for both modern and ancient fluvial systems. In addition, Fig. 17 shows the range of channel deposit dimensions interpreted from geologic data collected for the Trinity River levee project (i.e., Roig-Silva et al., 2010). Typically, the w/t ratio for channel deposits decreases with the stability of the river channel (Gouw and Berendsen, 2007), with high ratios associated with highly sinuous and braided channels and low ratios associated with incised or straight channels. The modeled values are in general agreement

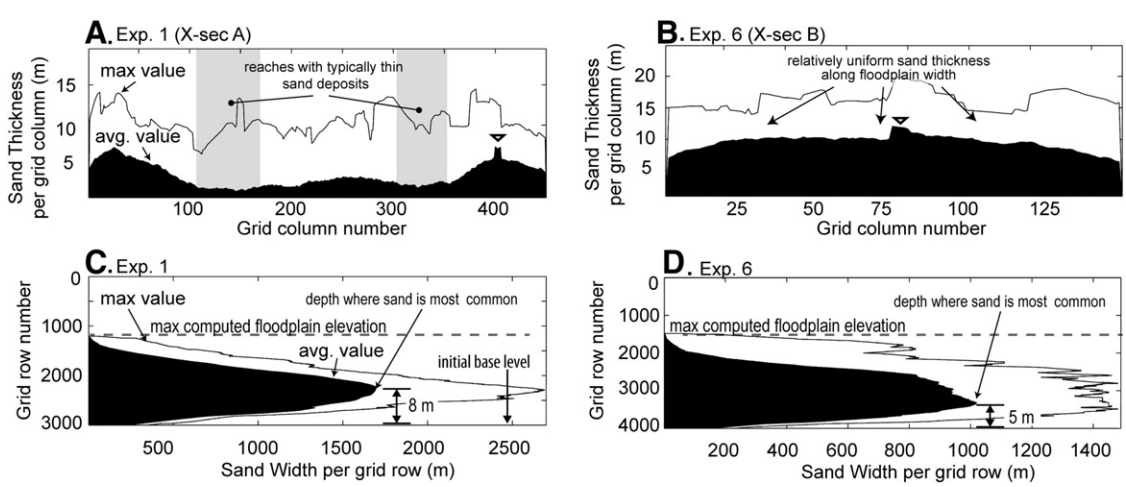

Fig. 15. [A, B] Average and maximum values predicted for the total sand thickness along the floodplain width per each column of the model's computational grid (for the two base experiments). Each grid column represents a 10-m segment of floodplain. [C, D] Average and maximum sand width within the floodplain per row of the model's computational grid. Each grid row represents a 0.01-m segment of floodplain depth. The open triangles show the lateral position of the channel along the floodplain width at the start of each model run.

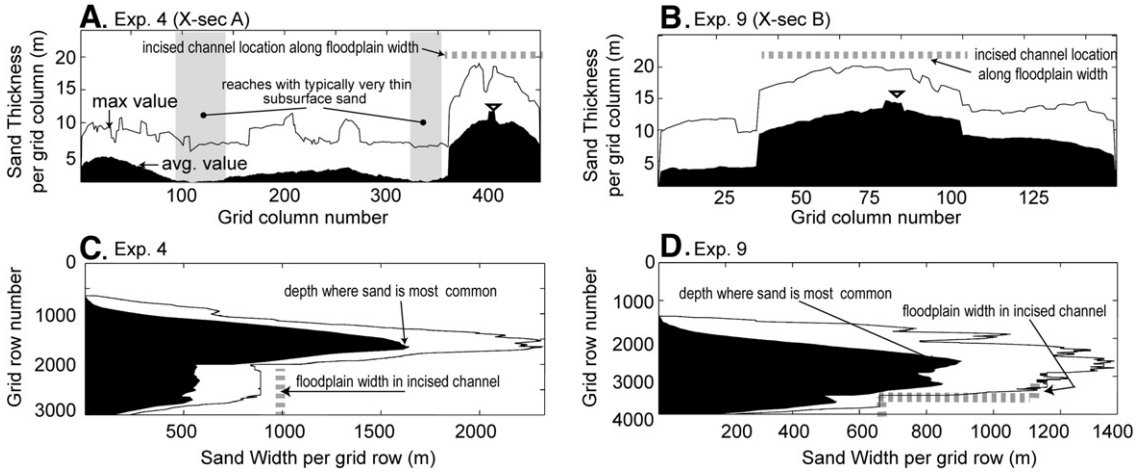

Fig. 16. [A, B] Average and maximum values predicted for the total sand thickness along the floodplain width per each column of the model's computational grid (for Experiments 4 and 9). Each grid column represents a 10-m segment of floodplain. [C, D] Average and maximum sand width within the floodplain per row of the model's computational grid. Each grid row represents a 0.01-m segment of floodplain depth. The open triangles show the lateral position of the channel along the floodplain width at the start of each model run.

with the data reported for fully meandering river channels (Fielding and Crane, 1987; Gibling, 2006). Relative to the full range of values reported for meandering channels, the variations in the distribution of channel deposit values due to the different experimental scenarios examined in this study are small. This could indicate that the employed model is not as sensitive to the range of analyzed parameters as natural fluvial systems, that there are additional influential processes that the model does not consider, or both. The distribution of channel deposit dimensions computed during an individual experiment is similar in size to the distribution of channel deposit dimensions measured along an individual river reach in large fluvial systems (e.g., Gouw and Berendsen, 2007).

5.3. Engineering significance

The presence of channel deposits within a floodplain is an important consideration for local engineering projects that may be adversely affected by seepage (USACE, 1956). Poorly graded sand has high saturated hydraulic conductivities between 10^{-5} and 10^{-4} m s^{-1} and is an effective medium for seepage. As the percentage of fines increases (i.e., silts and clays compose > 12%), the hydraulic conductivity decreases rapidly (i.e., it may drop below 10^{-6} m s^{-1}). In floodplain environments, poorly-graded sand is primarily a product of past river channel deposition (Fisk, 1945; Allan, 1974; Kolb, 1975). If a buried channel deposit is located underneath a dam or a levee, it could serve as a conduit for subsurface fluid seepage between the hydraulically-loaded side (i.e., the side impounding water) of the structure and the protected side (i.e., the landward side of levees, the downstream side of dams) of the structure (USACE, 1956; Turnbull and Mansur, 1959). Seepage of a sufficient magnitude is capable of entraining and transporting a significant amount of the material that it flows through (i.e., internal erosion or piping). This erosion can undermine the structural integrity of a dam or a levee causing it to subside or collapse, which can lead to structural failure. For this reason, any knowledge about the size and distributions of channel deposits within a river floodplain is extremely advantageous to engineers tasked with planning new structures or assessing the risk of failure for existing structures in floodplain environments.

Typically, engineers characterize floodplain geotechnical properties by relying on data collection methods such as geologic borings or cone penetrometer tests (CPTs). These types of methods collect discrete data points that require geostatistical techniques (e.g., kriging, inverse-distance weighting) to interpolate the derived geotechnical property data over a two or three-dimensional area (Myers, 1994). The accuracy of the interpolation depends on the balance between the overall size of the project area and the density at which the geotechnical property data are collected. It often is impossible to collect data over a project area at the resolution required to identify all of the transitions between areas of high-seepage potential (e.g., channel deposits) and low-seepage potential due to capital and time limitations. Boring and CPT plans often are based on generic grid patterns that are sometimes designed to increase in density around areas that are predicted to be at higher risk of seepage, such as former river channel locations. The locations of former river channels are assessed by referencing geomorphological maps (e.g., Saucier, 1994) and historical imagery; however, this information often has spatial (e.g., small map scale) or temporal resolutions (e.g., imagery from few time periods) that hinder its practical benefits. For example, data availability typically limits the period of time in which a channel location is identifiable to less than 100 years. In areas where the drainage patterns have remained relatively constant since the last climatic shift (≈10,000 yrs BP) and where the river channel positions are in constant flux due to meandering and floodplain

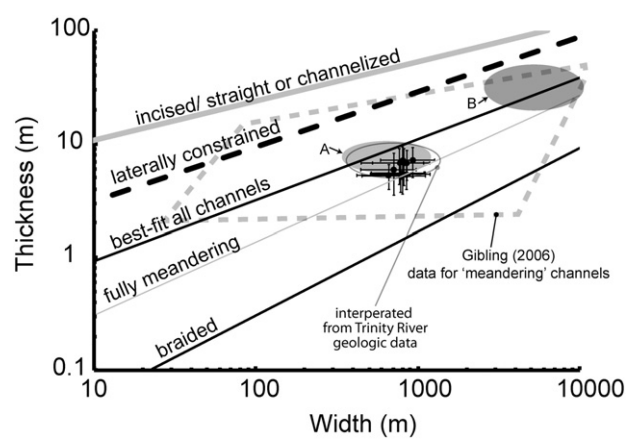

Fig. 17. The modeled channel deposit dimensions relative to values observed in nature. The black circles are the averaged dimension values for the 10 experiments. The error bars represent the standard deviation of the values in each experiment. The exponential functions represented by the five trend lines were derived by Fielding and Crane (1987) from channel and deposit data from forty-five published sources. The polygon bound by a dashed, gray line is the range of channel deposit dimensions found for meandering channels by Gibling (2006) who analyzed data from one thousand-five hundred different deposits. The shaded ovals approximately define the range of meander belt dimensions measured for two well studies river systems, (A) the Linge River within the Rhine–Meuse Delta, Netherlands and (B) the Lower Mississippi River, USA as reported in Gouw and Berendsen (2007). The open gray oval illustrates the range of deposit dimensions estimated from geologic cross-sections of the Upper Trinity River interpreted from Roig-Silva et al. (2010).

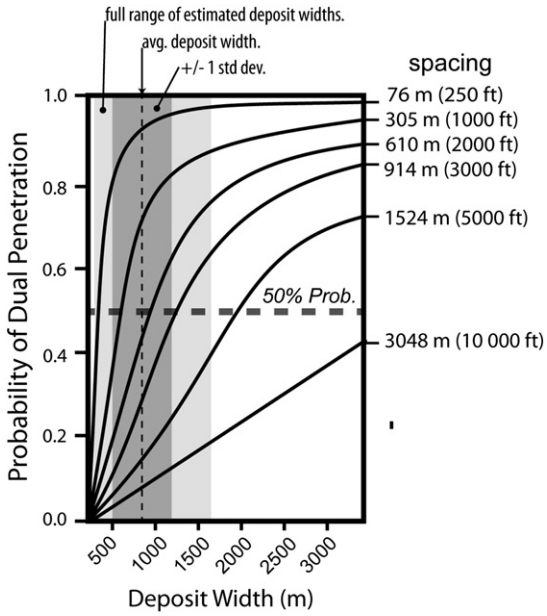

Fig. 18. The probability a channel deposit would be penetrated twice at specified boring spacings. This figure is based on the statistical model presented in Lorenz et al. (1985) and May and Schmitz (1996). The figure also shows channel width data modeled from the Trinity River project area at X-sec A.

deposition, approximately 1.0 percent of the channel deposits would be identifiable by traditional methods.

Alluvial architecture models, such as those discussed in this paper, cannot identify specific channel deposit locations or the precise dimensions of the channel deposits but, with minimal input data, can estimate the relative probability that an area contains channel deposits and the likely distribution of the channel deposit dimensions. This information is extremely useful to engineers for optimizing data collection methods to meet a prescribed balance between the risk of project failure and project cost. For example, Lorenz et al. (1985) derived a function that can be used to calculate the boring spacings required to effectively characterize the alluvial architecture of a project area if the ranges of channel deposit widths are known. The function assumes that the channel deposit has an infinite length and that effective characterization requires the deposit to be penetrated (i.e., sampled) twice and is based on the laws of compound probability and total probability (see May and Schmitz (1996) for a full description of the function and of its application). As the probability that a channel deposit is effectively characterized increases, the risk that a channel deposit will go undetected and undermine an engineering structure decreases; however, as the distance between boring spacings decreases, the project costs increase.

By employing the alluvial architecture model introduced in this study with the calibration data shown in Table 2, a distribution of probable channel deposit dimensions can be estimated. For the Trinity River project area, the channel deposit widths are estimated to range from a minimum of 60 m to a maximum near 1600 m, with the majority (i.e., +/− one standard deviation from the mean value) of deposits width ranging between 500 and 1170 m. Coupling the model predictions with the Lorenz et al. (1985) function allows a project engineer to associate alternative boring spacings with the probability that they will accurately characterize the geotechnical properties of the Trinity River floodplain (Figure 18). This information can be used to optimize boring plans by weighing the marginal costs associated with drilling additional borings against the desired level of certainty that the seepage risk is characterized.

For the range of modeled deposit widths in the Trinity River project area, a 76-m (250 ft) boring spacing would produce an 80–95% probability that the area subsurface would be effectively characterized.

Table 6
Average effect of the different experimental scenarios on model results relative to the 'base case' results unless otherwise stated. Values in italics indicate the scenario produced both increased and decreased values depending on the cross section location modeled.

Scenario	# Deposits	# Process avulsions	Width	Mean thickness
X-sec B vs. X-sec A	−30%	−59%	+10%	+26%
Dynamic climate	+16%	+25%	−8%	−4%
Double sed. rate	+53%	+148%	−18%	−1%
Complex initial topo.	+5%	−7%	−9%	*+2%*
DC and *CT*	+19%	*+6%*	−17%	*+2%*

Increasing the boring spacing to 3080 m (10,000 ft) reduces the probability that the area subsurface would be effectively characterized to 3–15%. If an engineer wished to exceed a 50% probability that the majority of channel deposits were effectively characterized, the model results suggest that the engineer should use at least a spacing of under 914 m (3000 ft). A boring plan using this spacing would identify the locations of the majority of channel deposits in the project area, which compose the most likely seepage pathways for groundwater flow. Once the location of the possible seepage pathways are identified from the acquired boring data, engineers could focus seepage mitigation strategies (e.g., install seepage cut-off walls, seepage-preventing berms, groundwater relief wells) within these areas to reduce the probability that seepage would harm the project structures (i.e., levees).

Table 6 summarizes the average effects of the different experimental scenarios on key model output parameters. Scenario components including the alluvial basin width, the initial topography of the floodplain base level, and the channel aggradation rate were found to significantly affect the fraction of the floodplain width that contained channel deposits. It is this fraction of the floodplain that would be most prone to channel deposit-related seepage. These scenario components primarily influenced the distribution of the channel deposits by impacting the channel avulsion frequency during the floodplain construction. Avulsion frequency was found to control the width and the total amount of the channel deposits within a floodplain cross section but in contrasting ways (i.e., increasing the total number of deposits typically decreased the deposit widths). This phenomenon made it difficult to identify how each experimental scenario explicitly affected the spatial distribution of the channel deposits.

6. Conclusions

[1] This study presents a simple computational model that predicts the amount, size of, and relative distribution of sedimentary channel deposits within a floodplain cross section based on a relatively small number of boundary conditions relating to geomorphic processes.

[2] The types of information provided by this modeling study are informative for technicians and planners working on engineering projects where very little information is known about the subsurface environment, which is often the case. The model does not provide deterministic estimates of seepage magnitude or location; rather it computes a distribution of possible frequencies and dimensions of seepage-prone areas based on a set of geomorphology-based rules. While this information is simplistic, it can be valuable for a number of tasks.

[3] Two cross sections were modeled for the Trinity River floodplain near Dallas, Texas. Five model scenarios were simulated in each cross section to explore how the scenario components influenced the channel deposit dimensions and the spatial distribution of the deposits. Alluvial basin width, the initial topography of the floodplain base level, and the channel aggradation rate were found to affect the fraction of the floodplain width that contained channel

deposits and, therefore, the fraction of the project area prone to channel deposit-related seepage.

Interpretation of the model results suggests that relatively high channel aggradation rates and wide floodplains increase the frequency of channel avulsions. As previously stated, channel avulsions lead to smaller but more numerous and spatially distributed channel deposits. From an engineering perspective, a floodplain constructed under a fluvial regime with a high avulsion frequency would have a higher degree of uncertainty surrounding the location of possible sand-related seepage sites. The smaller channel-deposit dimensions would decrease the probability that a single deposit fully traversed the subsurface area beneath a levee or a dam, a situation that puts that structure at greater risk of failure due to seepage.

[4] Alluvial architecture models, such as that employed in this study, can help produce an informed estimate of the distribution of sand within the floodplain. This information can be directly used to calculate the probability that a boring plan will effectively characterize the spatial distribution of geotechnical properties within a project area. Identification of channel-stacking patterns that selectively favor the deposition of channel sediment into certain reaches of the floodplain relative to other areas, such as those observed during the experiments located in X-sec A, could help to spatially differentiate the seepage risk within a floodplain.

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